

Week 4 - Quasiconvexity and quasiconcavity

Recall (Convex and concave functions). Let $X \subseteq \mathbb{R}^n$ be convex and let $f : X \rightarrow \mathbb{R}$.

- f is *convex* if for all $\mathbf{x}, \mathbf{y} \in X$ and all $\alpha \in [0, 1]$,

$$f(\alpha \mathbf{x} + (1 - \alpha) \mathbf{y}) \leq \alpha f(\mathbf{x}) + (1 - \alpha) f(\mathbf{y}).$$

- f is *concave* if for all $\mathbf{x}, \mathbf{y} \in X$ and all $\alpha \in [0, 1]$,

$$f(\alpha \mathbf{x} + (1 - \alpha) \mathbf{y}) \geq \alpha f(\mathbf{x}) + (1 - \alpha) f(\mathbf{y}).$$

Equivalently:

- f is convex $\iff$ its *epigraph*

$$\text{epi}(f) = \{(x, y) \in X \times \mathbb{R} : y \geq f(x)\},$$

is a convex set.

- f is concave $\iff$ its *hypograph* (a.k.a. subgraph)

$$\text{hypo}(f) = \{(x, y) \in X \times \mathbb{R} : y \leq f(x)\}.$$

is a convex set.

Definition (Quasiconvexity and quasiconcavity). Let $X \subseteq \mathbb{R}^n$ be convex and let $f : X \rightarrow \mathbb{R}$.

- (1) f is *quasiconvex* if for all $\mathbf{x}, \mathbf{y} \in X$ and all $\alpha \in [0, 1]$,

$$f(\alpha \mathbf{x} + (1 - \alpha) \mathbf{y}) \leq \max\{f(\mathbf{x}), f(\mathbf{y})\}.$$

- (2) f is *strictly quasiconvex* if for all $\mathbf{x} \neq \mathbf{y}$ in X and all $\alpha \in (0, 1)$,

$$f(\alpha \mathbf{x} + (1 - \alpha) \mathbf{y}) < \max\{f(\mathbf{x}), f(\mathbf{y})\}.$$

- (3) f is *quasiconcave* if for all $\mathbf{x}, \mathbf{y} \in X$ and all $\alpha \in [0, 1]$,

$$f(\alpha \mathbf{x} + (1 - \alpha) \mathbf{y}) \geq \min\{f(\mathbf{x}), f(\mathbf{y})\}.$$

- (4) f is *strictly quasiconcave* if for all $\mathbf{x} \neq \mathbf{y}$ in X and all $\alpha \in (0, 1)$,

$$f(\alpha \mathbf{x} + (1 - \alpha) \mathbf{y}) > \min\{f(\mathbf{x}), f(\mathbf{y})\}.$$

Proposition (Convex/concave $\Rightarrow$ quasi-(con)vex). Let $X \subseteq \mathbb{R}^n$ be convex and $f : X \rightarrow \mathbb{R}$.

- (1) If f is convex (resp. strictly convex), then f is quasiconvex (resp. strictly quasiconvex).

- (2) If f is concave (resp. strictly concave), then f is quasiconcave (resp. strictly quasiconcave).

Proof. (1) Fix $\mathbf{x}, \mathbf{y} \in X$ and $\alpha \in [0, 1]$, and set $\mathbf{z}_\alpha = \alpha\mathbf{x} + (1 - \alpha)\mathbf{y}$. By convexity,

$$f(\mathbf{z}_\alpha) \leq \alpha f(\mathbf{x}) + (1 - \alpha)f(\mathbf{y}) \leq \alpha \max\{f(\mathbf{x}), f(\mathbf{y})\} + (1 - \alpha) \max\{f(\mathbf{x}), f(\mathbf{y})\} = \max\{f(\mathbf{x}), f(\mathbf{y})\},$$

so f is quasiconvex. If f is strictly convex, then for $\mathbf{x} \neq \mathbf{y}$ and $\alpha \in (0, 1)$ we have

$$f(\mathbf{z}_\alpha) < \alpha f(\mathbf{x}) + (1 - \alpha)f(\mathbf{y}) \leq \max\{f(\mathbf{x}), f(\mathbf{y})\},$$

hence f is strictly quasiconvex.

(2) The concave case is analogous. Alternatively, apply part (1) to $-f$: if f is concave, then $-f$ is convex and thus quasiconvex, which is equivalent to f being quasiconcave. The strict variant follows in the same way. $\square$

Remark (Bounds along chords). If f is quasiconvex, then for every $x, y \in X$ and $\alpha \in [0, 1]$,

$$f(z_\alpha) \leq \max\{f(x), f(y)\}.$$

If f is quasiconcave, then

$$f(z_\alpha) \geq \min\{f(x), f(y)\}.$$

Thus, if f is both quasiconvex and quasiconcave,

$$\min\{f(x), f(y)\} \leq f(z_\alpha) \leq \max\{f(x), f(y)\}.$$

Under strict convexity/concavity, the inequalities are strict whenever $\mathbf{x} \neq \mathbf{y}$ and $\alpha \in (0, 1)$.

Remark (Converse fails). The converse of the previous proposition is not always true.

Example (Increasing functions are (strictly) quasi-convex and quasi-concave). Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be increasing (resp. strictly increasing). Then f is quasiconvex and quasiconcave (resp. strictly quasiconvex and strictly quasiconcave).

Proof. Fix $x, y \in \mathbb{R}$ with $x > y$ and let $\alpha \in (0, 1)$. Set $z_\alpha = \alpha x + (1 - \alpha)y$, so $x > z_\alpha > y$. By monotonicity,

$$f(x) \geq f(z_\alpha) \geq f(y) \quad (\text{resp. } f(x) > f(z_\alpha) > f(y) \text{ if } f \text{ is strictly increasing}).$$

Therefore, for all $\alpha \in (0, 1)$,

$$\max\{f(x), f(y)\} \geq f(z_\alpha) \geq \min\{f(x), f(y)\},$$

and in the strict case,

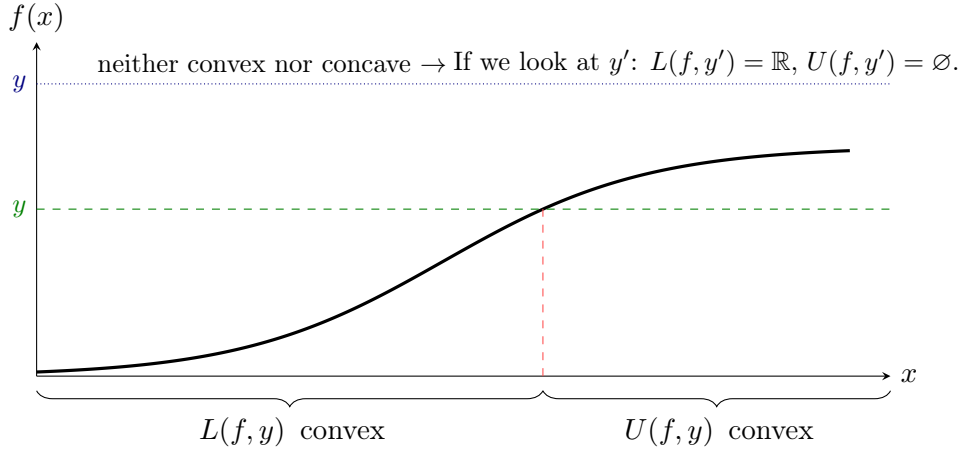
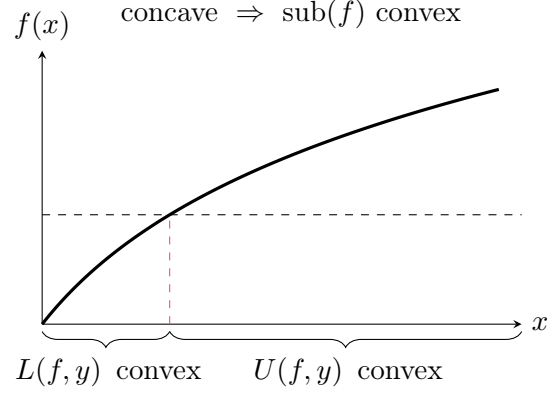
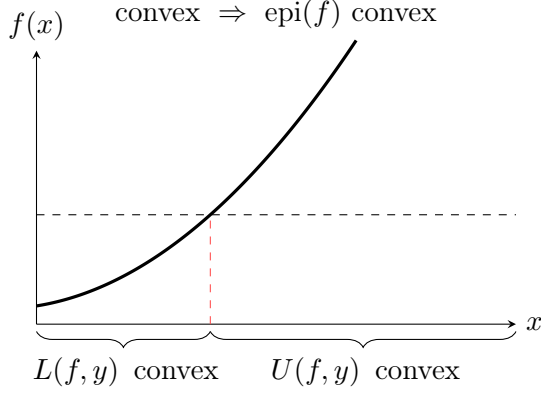
$$\max\{f(x), f(y)\} > f(z_\alpha) > \min\{f(x), f(y)\}.$$

These are precisely the (strict) quasiconvexity and (strict) quasiconcavity inequalities on $\mathbb{R}$.

Definition (Upper and lower contour sets). Let $X \subseteq \mathbb{R}^n$ be nonempty and let $f : X \rightarrow \mathbb{R}$. For any $y \in \mathbb{R}$ define

$$U(f, y) := \{x \in X : f(x) \geq y\} \subseteq X, \quad L(f, y) := \{x \in X : f(x) \leq y\} \subseteq X.$$

Here $U(f, y)$ is the set of inputs whose value is at least y (hypograph-like at level y), while $L(f, y)$ is the set of inputs whose value is at most y (epigraph-like at level y).



Proposition (Characterization by contour sets). *Let $X \subseteq \mathbb{R}^n$ be convex and $f : X \rightarrow \mathbb{R}$.*

(1) f is quasi-convex $\iff$ for all $y \in \mathbb{R}$ the lower boundary set

$$L(f, y) := \{x \in X : f(x) \leq y\}$$

is convex (possibly empty).

(2) f is quasiconcave $\iff$ for all $y \in \mathbb{R}$ the upper boundary set

$$U(f, y) := \{x \in X : f(x) \geq y\}$$

is convex (possibly empty).

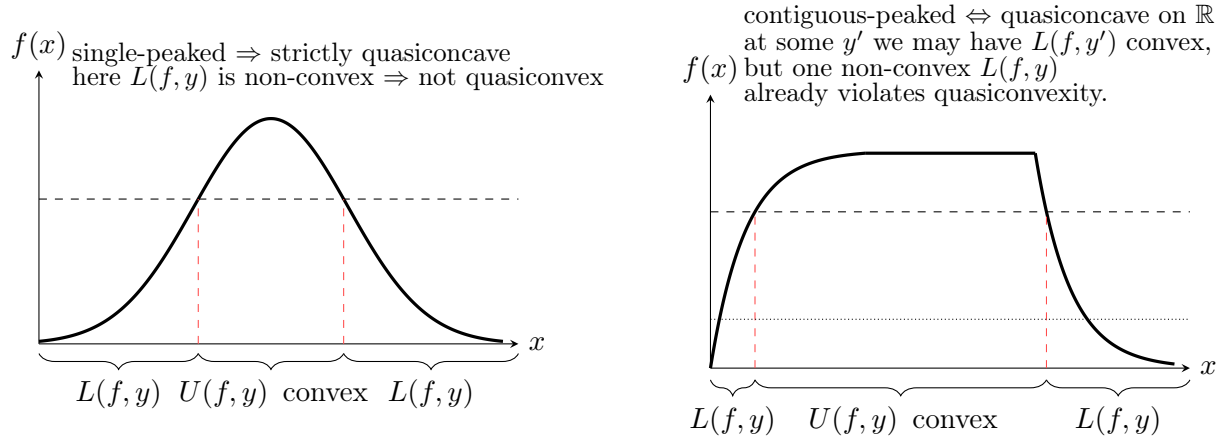
Remark (Clarification on “borderline cases”). At some levels y , it may happen that $L(f, y) = X$ or $U(f, y) = X$ (for example, if $y \geq \sup f(X)$, then $L(f, y) = X$, and if $y \leq \inf f(X)$, then $U(f, y) = X$), or that $L(f, y) = \emptyset$ or $U(f, y) = \emptyset$. All these cases satisfy the proposition because X (and, in particular, $\mathbb{R}$ when $X = \mathbb{R}$) is convex and the empty set is also convex.

Example (Single-peaked vs. contiguous-peaked on $\mathbb{R}$; quasi-concavity consequences). Let $f : \mathbb{R} \rightarrow \mathbb{R}$. We say that f is single-peaked if there exists $m \in \mathbb{R} \cup \{\pm\infty\}$ such that f is strictly increasing up to m and strictly decreasing after m . For finite m , equivalently, f is strictly increasing on $(-\infty, m]$ and strictly decreasing on $[m, \infty)$. We say that f is *contiguous-peaked* if each upper contour set

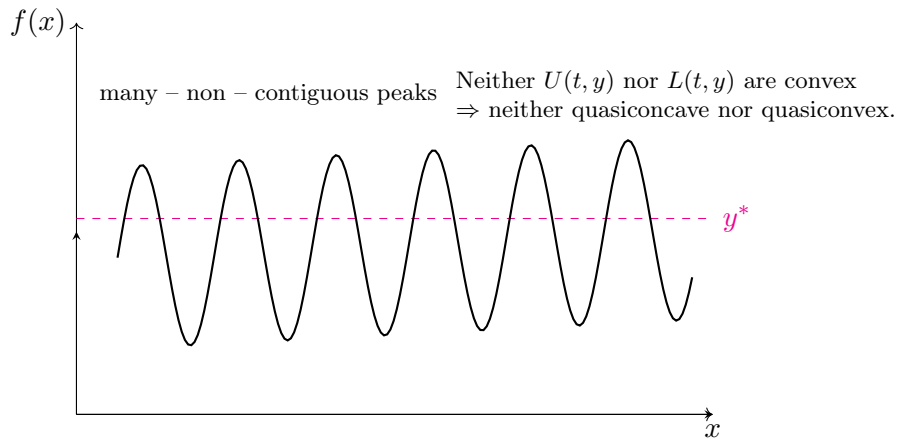
$U(f, y) = \{x \in \mathbb{R} : f(x) \geq y\}$ is an interval (possibly empty or all of $\mathbb{R}$), allowing for a flat “plateau” of maximizers.

Key facts (on $\mathbb{R}$).

- f is quasiconcave $\iff U(f, y)$ is an interval for every y (i.e., f is contiguous-peaked). Peaks may occur at $\pm\infty$ (monotone functions).
- If f is single-peaked, then f is *strictly* quasiconcave.



Remark (Why a single convex $L(f, y')$ does not suffice). Quasiconvexity requires *every* lower contour $L(f, y)$ to be convex. In the single-peaked panel, for the displayed y the set $L(f, y)$ is the union of two disjoint intervals (non-convex), so f fails quasiconvexity—even though for some other levels y' the set $L(f, y')$ happens to be convex.



Punchline. On $\mathbb{R}$:

- One contiguous peak (possibly a flat plateau) $\iff f$ is quasiconcave (equivalently, each $U(f, y)$ is an interval). The “peak” may occur at $+\infty$ or $-\infty$ (monotone functions).
- If f is single-peaked (strictly increasing up to a unique mode m and strictly decreasing thereafter—allowing $m = \pm\infty$), then f is *strictly* quasiconcave.

Level sets and level curves

Definition (Level set). Let $f : X \rightarrow \mathbb{R}$ and let $y \in \mathbb{R}$. The *level set* of f at level y is

$$C(f, y) := \{x \in X : f(x) = y\}.$$

Since

$$U(f, y) = \{x \in X : f(x) \geq y\}, \quad L(f, y) = \{x \in X : f(x) \leq y\},$$

we have

$$C(f, y) = U(f, y) \cap L(f, y).$$

Remark. When $X \subseteq \mathbb{R}^2$, a level set is often represented geometrically as a *level curve*. Thus, if

$$f(\tilde{x}_1, \tilde{x}_2) = f(\hat{x}_1, \hat{x}_2) = y,$$

then

$$(\tilde{x}_1, \tilde{x}_2), (\hat{x}_1, \hat{x}_2) \in C(f, y),$$

so the two points lie on the same level curve.

Intuition. A level curve collects all input vectors that generate exactly the same value of the function. In consumer theory, when f represents utility, these are the familiar *indifference curves*.

A two-dimensional single-peak example

Consider

$$f(x) = -\|x - c\|_2^2, \quad x \in \mathbb{R}^2,$$

where $c \in \mathbb{R}^2$ is fixed.

The function reaches its unique maximum at c , since

$$f(c) = 0$$

and

$$f(x) < 0 \quad \text{for every } x \neq c.$$

For any $y < 0$, the level set is

$$C(f, y) = \{x \in \mathbb{R}^2 : f(x) = y\} = \{x \in \mathbb{R}^2 : \|x - c\|_2 = \sqrt{-y}\},$$

which is a circle centered at c .

The corresponding upper contour set is

$$U(f, y) = \{x \in \mathbb{R}^2 : f(x) \geq y\} = \{x \in \mathbb{R}^2 : \|x - c\|_2 \leq \sqrt{-y}\},$$

which is the closed disk enclosed by that level curve. Since every Euclidean ball is convex, $U(f, y)$ is convex.

Remark. This example also illustrates quasiconcavity directly. Since every upper contour set $U(f, y)$ is convex, f is quasiconcave. In fact, $f(x) = -\|x - c\|_2^2$ is strictly concave and therefore strictly quasiconcave.

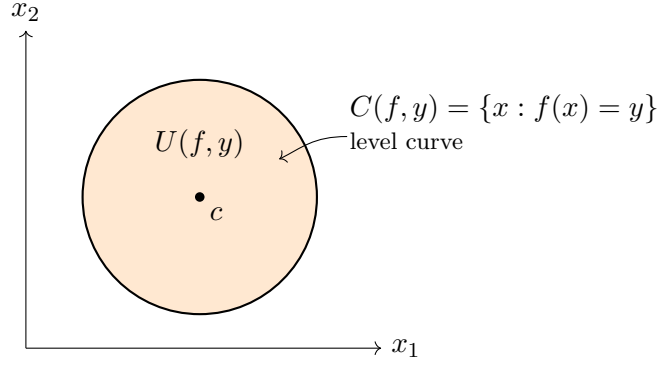


Figure 1: For a single-peaked radial function, the level curve is a circle and the corresponding upper contour set is the disk inside it.

Strictly monotone functions

Definition (Componentwise order). For $x, x' \in \mathbb{R}^n$, we write

$$x \geq x'$$

if

$$x_i \geq x'_i \quad \text{for every } i = 1, \dots, n.$$

Thus $x \geq x'$ means that x has at least as much of every coordinate as x' .

Definition (Monotonicity). Let $X \subseteq \mathbb{R}^n$.

A function $f : X \rightarrow \mathbb{R}$ is *monotone* or *weakly increasing* if

$$x \geq x' \implies f(x) \geq f(x').$$

It is *strictly monotone* if

$$x \geq x' \quad \text{and} \quad x \neq x' \implies f(x) > f(x').$$

Equivalently, if every coordinate of x is at least as large as the corresponding coordinate of x' and at least one coordinate is strictly larger, then $f(x) > f(x')$.

Intuition. Strict monotonicity says that moving weakly upward in every coordinate and strictly upward in at least one coordinate must strictly increase the value of the function. In consumer theory, this corresponds to the idea that “more is better”: if one bundle contains at least as much of every good and strictly more of at least one good, it must yield strictly higher utility.

Strict monotonicity and the shape of level curves

Proposition. Let $f : X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ be strictly monotone. If $x, x' \in X$ are distinct points on the same level set,

$$f(x) = f(x'),$$

then neither point can dominate the other componentwise.

Proof. Suppose, toward a contradiction, that

$$x \geq x' \quad \text{and} \quad x \neq x'.$$

Strict monotonicity would imply

$$f(x) > f(x'),$$

contradicting

$$f(x) = f(x').$$

The same argument rules out $x' \geq x$. □

In $\mathbb{R}^2$, this has an especially useful geometric implication.

Corollary. Let $f : X \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$ be strictly monotone, and suppose that

$$(\tilde{x}_1, \tilde{x}_2), (\hat{x}_1, \hat{x}_2) \in C(f, y)$$

are distinct points on the same level curve. If

$$\hat{x}_1 > \tilde{x}_1,$$

then necessarily

$$\hat{x}_2 < \tilde{x}_2.$$

Proof. Suppose instead that

$$\hat{x}_2 \geq \tilde{x}_2.$$

Then

$$(\hat{x}_1, \hat{x}_2) \geq (\tilde{x}_1, \tilde{x}_2)$$

componentwise, with strict inequality in at least the first coordinate. Strict monotonicity would therefore imply

$$f(\hat{x}_1, \hat{x}_2) > f(\tilde{x}_1, \tilde{x}_2),$$

contradicting the fact that the two points belong to the same level set. □

Intuition. For a strictly monotone function of two variables, two distinct points on the same level curve must involve a trade-off. If x_1 increases, x_2 must decrease in order to keep the value of f unchanged.

Thus, whenever the level set can be represented as a smooth curve, it has the familiar downward-sloping shape. This conclusion follows from monotonicity alone; differentiability is not required.

Monotonicity and upper contour sets

Proposition. Let $f : X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ be monotone. For every $y \in \mathbb{R}$, its upper contour set

$$U(f, y) = \{x \in X : f(x) \geq y\}$$

is upward closed with respect to the componentwise order. That is,

$$x \in U(f, y), \quad x' \geq x \quad \implies \quad x' \in U(f, y),$$

provided $x' \in X$.

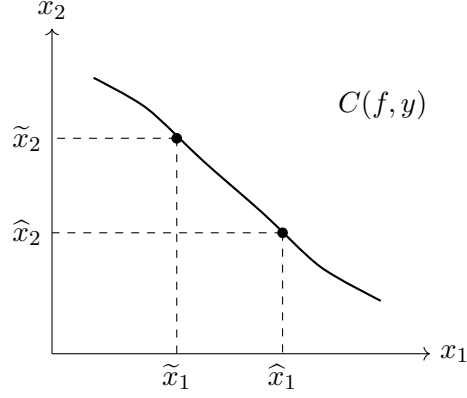


Figure 2: Under strict monotonicity, two distinct points on the same level curve cannot be componentwise ordered. Hence a higher x_1 must be accompanied by a lower x_2 .

Proof. If $x \in U(f, y)$, then

$$f(x) \geq y.$$

If $x' \geq x$, monotonicity implies

$$f(x') \geq f(x) \geq y.$$

Hence

$$x' \in U(f, y).$$

□

Remark. If f is strictly monotone and $x' \geq x$ with $x' \neq x$, then

$$f(x') > f(x).$$

Thus moving northeast from a point on a level curve moves strictly into a higher contour set.

Monotonicity versus quasiconcavity

It is important to distinguish two different geometric properties.

- **Monotonicity** determines which direction increases the value of the function. In particular, upper contour sets are upward closed.
- **Quasiconcavity** determines the shape of the upper contour sets. Specifically,

$$f \text{ is quasiconcave} \iff U(f, y) \text{ is convex for every } y \in \mathbb{R}.$$

Therefore, strict monotonicity by itself does *not* imply quasiconcavity.

Remark (Level curves versus contour sets). Quasiconcavity does not require the level curve

$$C(f, y) = \{x : f(x) = y\}$$

itself to be a convex set. Rather, it requires the entire region

$$U(f, y) = \{x : f(x) \geq y\}$$

lying weakly above that level to be convex.

Equivalently, if $x, x' \in U(f, y)$, quasiconcavity requires that every mixture

$$\alpha x + (1 - \alpha)x', \quad \alpha \in [0, 1],$$

also belong to $U(f, y)$.

Since

$$f(x) \geq y, \quad f(x') \geq y,$$

quasiconcavity guarantees

$$f(\alpha x + (1 - \alpha)x') \geq \min\{f(x), f(x')\} \geq y.$$

If f is *strictly quasiconcave*, then for distinct x, x' and $\alpha \in (0, 1)$,

$$f(\alpha x + (1 - \alpha)x') > \min\{f(x), f(x')\} \geq y.$$

Two examples: convex and non-convex upper contour sets

Example (Strictly monotone and quasiconcave). Let

$$X = \mathbb{R}_{++}^2$$

and define

$$f(x_1, x_2) = \log x_1 + \log x_2.$$

The function is strictly monotone because

$$\frac{\partial f}{\partial x_1} = \frac{1}{x_1} > 0, \quad \frac{\partial f}{\partial x_2} = \frac{1}{x_2} > 0.$$

For a given $y \in \mathbb{R}$,

$$U(f, y) = \{(x_1, x_2) \in \mathbb{R}_{++}^2 : \log x_1 + \log x_2 \geq y\}.$$

Equivalently,

$$x_1 x_2 \geq e^y,$$

or

$$x_2 \geq \frac{e^y}{x_1}.$$

Since the function

$$x_1 \mapsto \frac{e^y}{x_1}$$

is convex on $\mathbb{R}_{++}$, $U(f, y)$ is its epigraph and is therefore convex. Hence f is quasiconcave. In fact, f is strictly concave and therefore strictly quasiconcave.

Example (Strict monotonicity does not imply quasiconcavity). Let

$$X = \mathbb{R}_+^2$$

and define

$$f(x_1, x_2) = x_1^2 + x_2^2.$$

On $\mathbb{R}_+^2$, f is strictly monotone in the componentwise order.

For $y > 0$,

$$U(f, y) = \{(x_1, x_2) \in \mathbb{R}_+^2 : x_1^2 + x_2^2 \geq y\}.$$

This upper contour set is not convex.

Indeed, consider

$$x = (\sqrt{y}, 0), \quad x' = (0, \sqrt{y}).$$

Both satisfy

$$f(x) = f(x') = y,$$

so

$$x, x' \in U(f, y).$$

However, their midpoint is

$$\frac{x + x'}{2} = \left(\frac{\sqrt{y}}{2}, \frac{\sqrt{y}}{2} \right),$$

and

$$f\left(\frac{x + x'}{2}\right) = \frac{y}{4} + \frac{y}{4} = \frac{y}{2} < y.$$

Thus

$$\frac{x + x'}{2} \notin U(f, y),$$

so $U(f, y)$ is not convex and f is not quasiconcave.

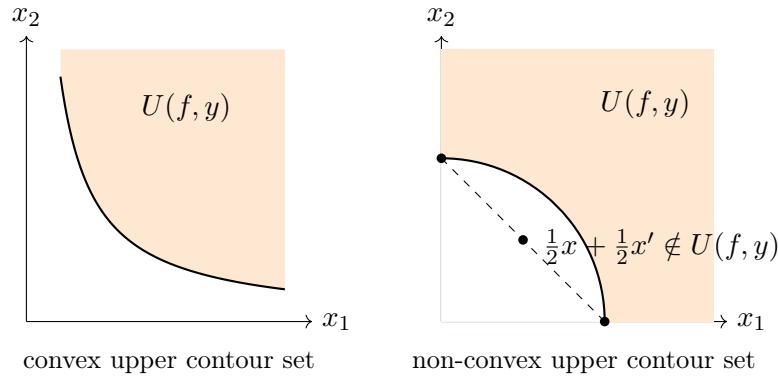


Figure 3: Strict monotonicity does not determine whether upper contour sets are convex. Quasi-concavity requires convexity of every upper contour set.

Economic interpretation

If f represents preferences over bundles, then:

- a level curve

$$C(f, y) = \{x : f(x) = y\}$$

contains bundles that give the same utility;

- strict monotonicity implies that two distinct bundles on the same level curve cannot dominate one another componentwise;

- the upper contour set

$$U(f, y) = \{x : f(x) \geq y\}$$

contains all bundles that are weakly preferred to the level y (this will appear in Kreps as the 'No Worse Than' set, NWT);

- quasiconcavity means that if two bundles are weakly preferred to a given benchmark, then every convex combination of those bundles is also weakly preferred to that benchmark.

This is the geometric link between quasiconcavity of a utility representation and convexity of upper contour sets.